

Foreign-Exchange Volatility, Risk and Trend Strategies

A reproducible study of EUR/USD, GBP/USD and USD/JPY (2022–2024)

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Abstract

We ask one question of three FX majors over 2022–2024: *does conditioning risk on time-varying, fat-tailed volatility improve real risk decisions — VaR/ES coverage and exception independence — or is a constant-variance model good enough?* We establish fat tails and clustering (diagnostics), fit EWMA and GARCH(1,1) Student- t volatility, and push the conditional volatility into **conditional VaR/ES** (GARCH-filtered and Filtered Historical Simulation) backtested side-by-side against the unconditional model (Kupiec, Christoffersen), validate the Expected Shortfall (Acerbi-Székely) and model the regulatory tail by **Extreme Value Theory**, and finally let the vol model *size* a momentum strategy whose Sharpe carries bootstrap confidence intervals. The honest finding: conditioning improves exception independence for the European pairs, while for USD/JPY parsimony is hard to beat and the GARCH- t ES is rejected. A methodological cornerstone: weekend/holiday dates are *non-observations*, not missing data — a naive calendar imputation fabricates roughly a quarter of the sample, so the study runs on the native 781 trading days. Every number is produced by a unit-tested Python package and reproduces from a committed snapshot.

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1 Introduction

Volatility is the central object of FX risk management: it drives position sizing, option pricing and Value-at-Risk. This report takes three liquid currency pairs — EUR/USD, GBP/USD and USD/JPY — and carries volatility from a descriptive rolling estimate through to a fitted conditional model, a backtested risk measure, and an out-of-sample forecast evaluation. The guiding principles are *honest data*, *tested computation*, and *claims traceable to printed values*: every figure and table below is generated by the accompanying `fxvol` package and reproduces offline from a committed snapshot. The study addresses four questions: (i) what are the distributional and time-series properties of daily FX returns; (ii) does conditional-volatility modelling (GARCH) add value over simpler baselines; (iii) are standard risk measures well calibrated; and (iv) does a simple trend rule survive realistic transaction costs?

2 Data and methodology

2.1 Data

Daily OHLC quotes for the three pairs are retrieved from Yahoo Finance for 2022-01-03–2024-12-30 and stored as a committed Parquet snapshot (as-of 2026-06-14) so the analysis is deterministic and network-free. Each pair has 781 trading observations.

2.2 The weekend-imputation pitfall

FX trades Monday–Friday; Saturday and Sunday are *non-observations*, not gaps to be filled. An earlier version of this project reindexed each series onto a full 365-day calendar and imputed the missing weekend/holiday rows with a recursive five-day rolling mean. Because the recursion feeds its own imputed values back in, it is a low-pass filter that **fabricated 312 of 1093 calendar rows ($\approx 28\%$)**. This contaminates every return, volatility, RSI and distributional statistic, induces artificial autocorrelation, and—critically—breaks the annualization, since $\sqrt{252}$ assumes 252 observations per year while a calendar series has ≈ 365 . We therefore discard the calendar reindex entirely and compute all statistics on the native trading-day index; a unit test guarantees no weekend row can re-enter the data. Table 1 reports the integrity check.

Table 1: Data-integrity check after trading-day alignment.

	Value
Trading rows / pair	781
Weekend rows	0
First date	2022-01-03
Last date	2024-12-30
Calendar days spanned	1093

2.3 Returns, volatility and indicators

We use log returns $r_t = \ln(P_t/P_{t-1})$ (time-additive). Annualized rolling volatility over a window of n trading days is $\hat{\sigma}_t^{\text{ann}} = \sqrt{252} \text{sd}(r_{t-n+1}, \dots, r_t)$, with the factor matched to the trading-day frequency. As a decay-weighted alternative we use the RiskMetrics EWMA recursion $\sigma_t^2 = (1 - \lambda)r_t^2 + \lambda\sigma_{t-1}^2$ with $\lambda = 0.94$. Trend and momentum are summarised by simple moving averages and Wilder’s RSI (recursive smoothing with $\alpha = 1/14$), the canonical definition.

Table 2: Daily log-return summary statistics (%).

	Mean	Std	Min	Q25	Median	Q75	Max
EURUSD	-0.011	0.496	-1.859	-0.296	-0.013	0.279	1.821
GBPUSD	-0.009	0.581	-4.232	-0.309	0.004	0.293	3.031
USDJPY	0.04	0.666	-3.796	-0.284	0.086	0.392	2.503

USD/JPY carries the only positive average drift, reflecting its sustained 2022–2024 appreciation, while GBP/USD exhibits the widest range (Table 2), a footprint of the late-2022 UK fiscal episode.

2.4 Visual overview

Figure 1 gives a per-pair overview: the closing price with its 20/50-day moving averages, the annualized rolling volatility, the daily-return distribution, and Wilder’s RSI.

3 Distributional and econometric diagnostics

Before modelling we test the properties that justify the models. Table 3 reports skewness and excess kurtosis, the Jarque–Bera (JB) normality test, ADF and KPSS stationarity tests, and Ljung–Box (LB) and Engle ARCH-LM tests for autocorrelation and conditional heteroskedasticity.

Table 3: Distributional and time-series diagnostics (p -values; LB at 10 lags).

	Skew	Exc kurt	JB p	ADF ret p	KPSS ret p	LB ret p	LB sq p	ARCH-LM p	ADF price p
EURUSD	-0.003	1.173	<0.001	<0.001	0.100	0.218	<0.001	<0.001	0.137
GBPUSD	-0.265	5.543	<0.001	<0.001	0.100	0.151	<0.001	<0.001	0.143
USDJPY	-0.516	2.823	<0.001	<0.001	0.100	0.599	0.427	0.576	0.284

Three facts emerge. (i) **Normality is rejected** for all pairs (JB $p < 0.001$); returns are leptokurtic, with excess kurtosis up to 5.5 for GBP/USD, visible in the QQ plot (Figure 2). (ii) Returns are **stationary** (ADF rejects a unit root, KPSS does not), whereas log-prices are not—the standard $I(1)$ price / $I(0)$ return picture. (iii) **Volatility clustering** is present for the European pairs

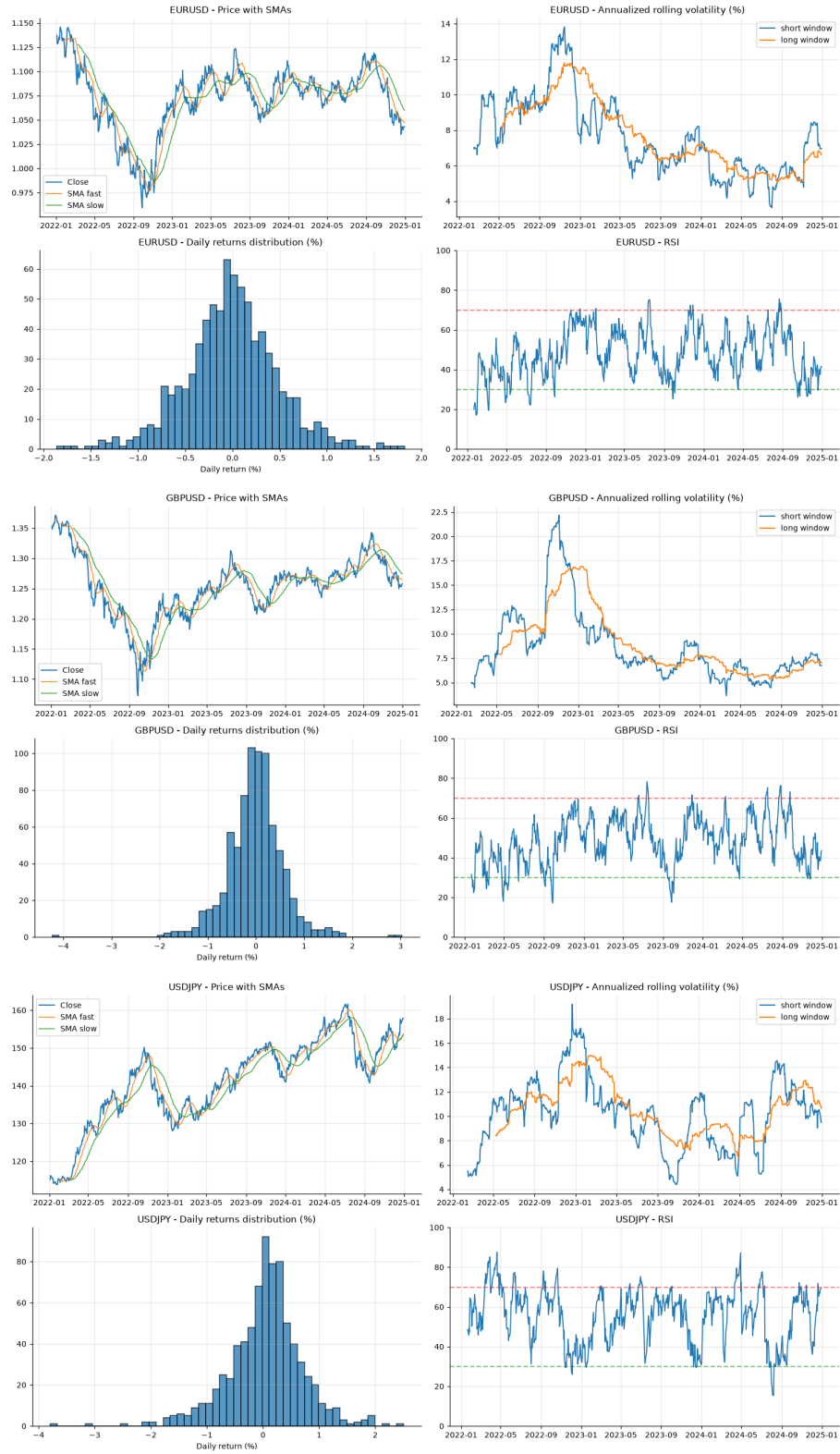


Figure 1: Per-pair dashboards: price with SMAs (top-left), annualized rolling volatility (top-right), daily-return histogram (bottom-left) and Wilder RSI (bottom-right).

(squared-return LB and ARCH-LM $p < 0.001$), motivating GARCH; for USD/JPY the in-sample ARCH effect is weak ($p = 0.58$), a nuance we carry through to the forecasting results.

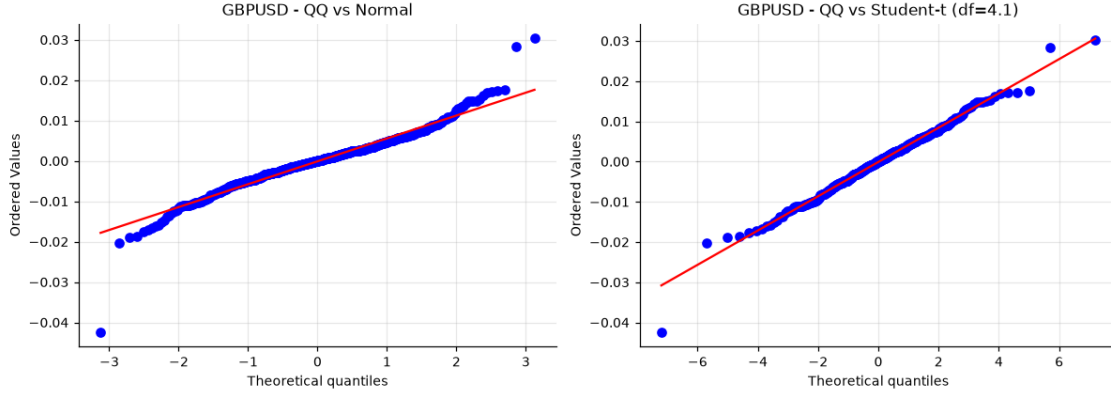


Figure 2: GBP/USD return quantiles against the Normal and a fitted Student- t . The heavy tails are evident in the departures from the Normal line.

4 Conditional-volatility modelling

We fit a GARCH(1,1) with Student- t innovations to each pair, $\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, where persistence is $\alpha + \beta$ and the unconditional (long-run) variance is $\omega/(1 - \alpha - \beta)$. Table 4 reports the estimates.

Table 4: GARCH(1,1) Student- t estimates (returns scaled $\times 100$ for fitting).

	omega	alpha	beta	nu (dof)	persistence	Long-run vol (%)
EURUSD	0.0006	0.0264	0.9715	7.7465	0.9979	8.6348
GBPUSD	0.0038	0.0488	0.9394	6.9151	0.9882	9.0531
USDJPY	0.0096	0.0723	0.9157	4.2933	0.988	14.1895

Persistence exceeds 0.98 for every pair—typical of daily FX, where volatility shocks decay slowly—and the estimated degrees of freedom (4.3–7.7) confirm fat-tailed innovations. Figure 3 overlays the GARCH conditional volatility on the rolling realized volatility and the EWMA estimate.

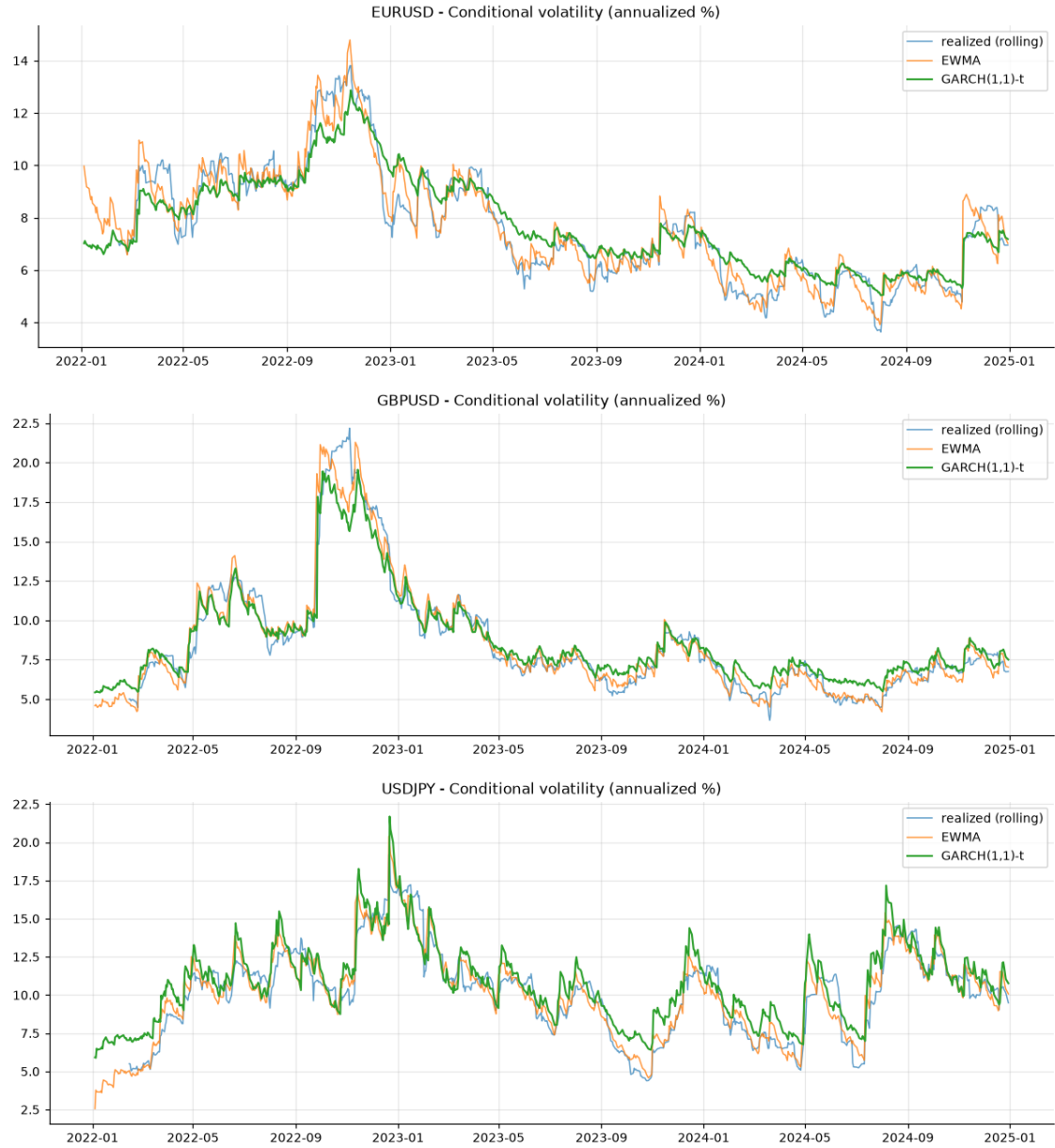


Figure 3: Annualized conditional volatility (GARCH(1,1)- t), EWMA and rolling realized volatility.

5 Risk measurement

For each pair we report one-day Value-at-Risk (VaR) and Expected Shortfall (ES) at the 95% and 99% levels, computed both historically and from a fitted Student- t , alongside annualized volatility, Sharpe and Sortino ratios, and maximum drawdown (Tables 5–6). VaR/ES are reported as positive loss fractions.

Table 5: Risk summary: volatility, risk-adjusted return, drawdown and shape.

	Ann vol	Sharpe	Sortino	Max DD	DD dur	Skew	Exc kurt
EURUSD	0.0788	-0.3553	-0.3543	-0.1651	184	-0.0031	1.1726
GBPUSD	0.0922	-0.2534	-0.2446	-0.221	184	-0.2653	5.543
USDJPY	0.1057	0.965	0.8804	-0.1503	62	-0.5155	2.8228

Table 6: One-day VaR and Expected Shortfall (loss fraction). H = historical, t = Student- t .

	VaR H 95	ES H 95	VaR t 95	ES t 95	VaR H 99	ES H 99	VaR t 99	ES t 99
EURUSD	0.0081	0.0112	0.0081	0.0112	0.0131	0.0156	0.013	0.0167
GBPUSD	0.0097	0.0135	0.0089	0.0133	0.0159	0.021	0.0155	0.0215
USDJPY	0.0109	0.0159	0.0097	0.0151	0.0174	0.0241	0.0178	0.0255

5.1 VaR backtesting

We roll a 250-day historical 99% VaR and test the breach sequence with the Kupiec proportion-of-failures test (unconditional coverage) and the Christoffersen test (conditional coverage). All three pass at the 5% level (Table 7), though USD/JPY is the weakest — 9 breaches vs 5.3 expected, Kupiec $p = 0.142$, Christoffersen $p = 0.112$ — foreshadowing the Expected-Shortfall rejection in §10.

Table 7: VaR backtest at 99% (rolling 250-day historical VaR).

	Breaches	Expected	Kupiec p	Christ. cc p
EURUSD	4	5.3	0.553	0.813
GBPUSD	6	5.3	0.765	0.893
USDJPY	9	5.3	0.142	0.112

6 Cross-pair dependence

Table 8 reports the return correlation matrix and Table 9 the principal-component decomposition. EUR/USD and GBP/USD co-move strongly (+0.78) and both move opposite USD/JPY, consistent with a dominant USD factor. The first principal component explains **70.7%** of joint variance, so diversification across these three majors is limited. Figure 4 shows the correlation heatmap and the 60-day rolling correlations, which widen markedly in stress periods.

Table 8: Return correlation matrix.

	EURUSD	GBPUSD	USDJPY
EURUSD	1	0.779	-0.435
GBPUSD	0.779	1	-0.439
USDJPY	-0.435	-0.439	1

Table 9: First principal-component loadings (common USD factor).

	PC1 loading
EURUSD	-0.619
GBPUSD	-0.62
USDJPY	0.483

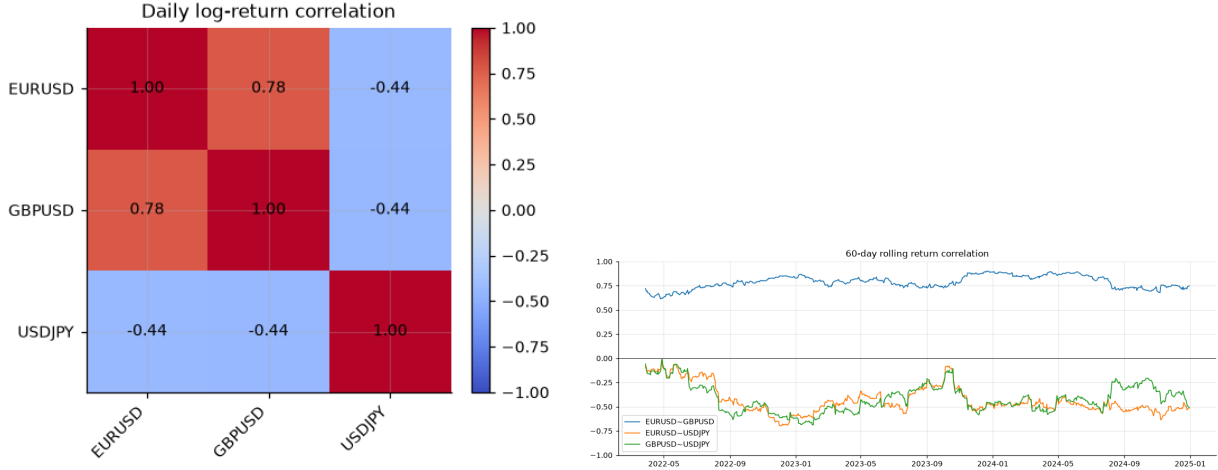


Figure 4: Left: return correlation heatmap. Right: 60-day rolling correlations.

7 Trend strategy backtest

We backtest a long/flat SMA(20/50) crossover. Signals are lagged one bar (next-bar execution, no look-ahead), and a 1.0 bp cost is charged on turnover. Performance is benchmarked against buy-and-hold (Table 10).

Table 10: SMA crossover performance vs buy-and-hold (B&H).

	Total ret	CAGR	Ann vol	Sharpe	Sortino	Max DD	Turnover	Hit rate	B&H ret	B&H Sharpe
EURUSD	0.0216	0.0069	0.045	0.1757	0.1303	-0.0821	5.1692	0.4459	-0.083	-0.316
GBPUSD	0.05	0.0159	0.0549	0.3144	0.2176	-0.0408	4.5231	0.5209	-0.0698	-0.2075
USDJPY	0.2572	0.0768	0.0812	0.9511	0.7347	-0.0945	4.8462	0.5742	0.3711	1.0192

The rule improves the Sharpe ratio over buy-and-hold for **EURUSD**, **GBPUSD**—by stepping aside during the 2022–2023 downtrends where holding lost money—but it does *not* beat USD/JPY’s persistent uptrend once costs are paid. We deliberately use a single textbook parameterization (no grid search), so in-sample optimism is minimal. Figure 5 shows the equity curves.

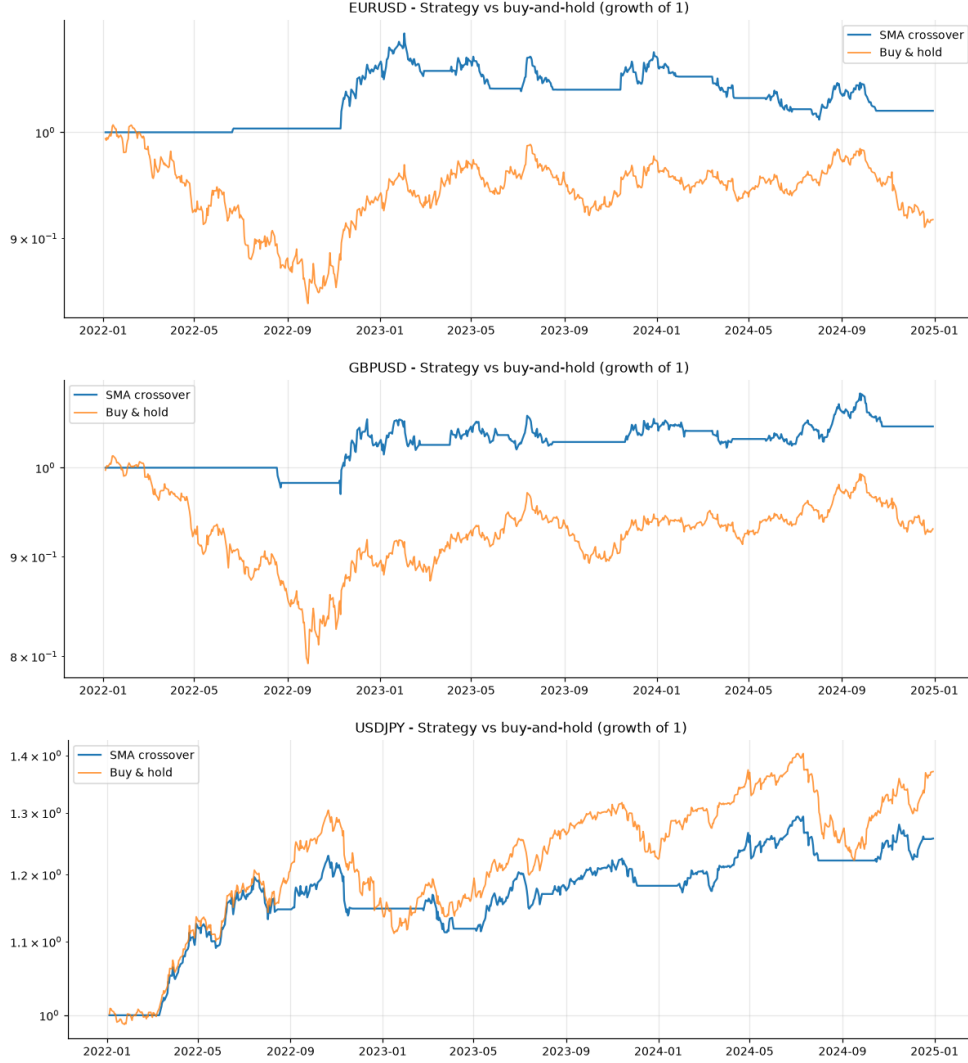


Figure 5: Strategy vs buy-and-hold growth of 1 (log scale).

8 Out-of-sample volatility forecasting

Finally we ask whether conditional-volatility modelling pays off out-of-sample. Using an expanding window, each model issues a one-day-ahead variance forecast; we score them with the QLIKE loss $L(\sigma^2, \hat{\sigma}^2) = \sigma^2 / \hat{\sigma}^2 - \ln(\sigma^2 / \hat{\sigma}^2) - 1$ against the squared-return proxy, and test differences with the Diebold–Mariano (DM) statistic (negative = GARCH better). The “naive” benchmark is the expanding constant variance (Table 11).

Table 11: Out-of-sample mean QLIKE per model and DM test of GARCH vs the naive benchmark.

	GARCH	EWMA	Rolling	Naive	Best	DM vs naive	p
EURUSD	1.57	1.606	1.721	1.665	garch	-1.3	0.195
GBPUSD	1.565	1.556	1.637	1.769	ewma	-4.46	0
USDJPY	1.98	1.995	2.208	1.915	naive	1.03	0.302

GARCH and EWMA beat the constant-variance benchmark on the European pairs (the DM difference is significant at 5% for GBPUSD). For **USD/JPY the naive benchmark wins**—consistent with its weak in-sample ARCH effect (§3): when conditional heteroskedasticity is faint,

a constant-variance forecast is hard to beat.

9 Conditional vs unconditional risk — the research spine

We now answer the central question. The GARCH conditional volatility is pushed into a one-day **conditional VaR/ES**: a GARCH-filtered parametric Student- t and a Filtered Historical Simulation (FHS) on standardized residuals. These are backtested against the rolling unconditional historical and Student- t VaR on identical out-of-sample dates. Table 12 shows the 95% Kupiec coverage p -values: the unconditional Student- t VaR is mis-calibrated (too few breaches), while the conditional models are well calibrated. Table 13 shows the Christoffersen *independence* p -values at 99%: conditioning reduces the breach clustering that static VaR suffers in the 2022 stress.

Table 12: Kupiec coverage p -values at 95% VaR (rows: model, columns: pair).

	EURUSD	GBPUSD	USDJPY
hist	0.26	0.118	0.26
t	0.013	0.044	0.103
garch_t	0.361	0.18	0.376
fhs	0.482	0.18	0.759

Table 13: Christoffersen independence p -values at 99% VaR (rows: model, columns: pair).

	EURUSD	GBPUSD	USDJPY
hist	0.805	0.71	0.118
t	0.853	0.853	0.599
garch_t	0.805	0.853	0.195
fhs	0.805	0.902	0.291

10 Expected Shortfall backtest and EVT tails

ES — not VaR — is the FRTB/Basel measure, so we validate the same GARCH-filtered Student- t ES of §9 with the Acerbi–Székely (2014) Z_2 test ($E[Z_2] = 0$ under correct ES; $Z_2 < 0$ means tail risk is understated), with a p -value simulated from the model’s own predictive law (Table 14). The ES is accepted for the European pairs but **rejected for USD/JPY** — the conditional model understates its tail, exactly the pair with weak in-sample ARCH. This motivates an Extreme-Value tail: a Generalized Pareto (Peaks-Over-Threshold, McNeil–Frey) fit to the standardized-residual tail (Table 15); the positive shape ξ confirms heavier-than-exponential tails the single Student- t cannot fully capture.

Table 14: Acerbi–Székely Z_2 ES backtest of the GARCH-filtered Student- t model at 99%.

	Z_2	p -value	breaches	expected
EURUSD	0.101	0.555	4	5.3
GBPUSD	0.497	0.876	3	5.3
USDJPY	-1.147	0.016	11	5.3

Table 15: POT/GPD tail of GARCH-standardized residuals (standardized scale).

	shape xi	n_exceed	POT VaR99 (std)	POT ES99 (std)
EURUSD	0.119	78	2.6	3.36
GBPUSD	0.074	78	2.86	3.69
USDJPY	0.063	78	2.71	3.47

11 Strategy with error bars

Finally the volatility model is put to work *sizing* risk: a time-series-momentum signal scaled to a 10% annualized vol target via the EWMA forecast, compared with the SMA crossover. Crucially every Sharpe is reported with a stationary-block-bootstrap 95% confidence interval and a Probabilistic Sharpe Ratio (Table 16). On ~ 750 daily observations **every bootstrap CI straddles zero, including USD/JPY** ($[-0.15, 2.08]$) — so no Sharpe is significant at 95%; only USD/JPY reaches a Probabilistic Sharpe Ratio above 0.95, and that rests on a single persistent trend.

Table 16: Strategy Sharpe with bootstrap 95% CI and Probabilistic Sharpe Ratio (PSR).

pair / strategy	Sharpe	CI low	CI high	PSR
EURUSD SMA(20/50)	0.176	-0.93	1.25	0.622
EURUSD TSMOM vol-target	-0.103	-1.13	0.91	0.429
GBPUSD SMA(20/50)	0.314	-0.63	1.27	0.71
GBPUSD TSMOM vol-target	0.409	-0.61	1.41	0.764
USDJPY SMA(20/50)	0.951	-0.15	2.08	0.951
USDJPY TSMOM vol-target	0.956	-0.16	2.05	0.953

Figure 6 shows the volatility-targeted equity curves.



Figure 6: Volatility-targeted TSMOM strategy vs buy-and-hold (growth of 1, log scale).

12 Discussion and limitations

The results cohere into one answer: conditioning FX risk on time-varying, fat-tailed volatility improves VaR exception *independence* and gives well-calibrated ES *where clustering is strong* (the European pairs); where in-sample ARCH is weak (USD/JPY) the parsimonious model is competitive and the GARCH- t ES is even rejected, which the EVT tail addresses. Strategy edges, honestly error-barred, are mostly indistinguishable from luck on this sample. Limitations: daily data over three years; a noisy realized-variance proxy (mitigated by reporting range-based estimators); a single, deliberately un-optimized strategy; and three pairs without asymmetric (GJR/EGARCH) or multivariate (DCC) volatility — deliberately scoped out as low signal-to-effort on this dataset. These are natural extensions.

13 Conclusion

On honest, weekend-free data, conditional-volatility modelling and properly backtested risk measures add demonstrable value, while a simple trend strategy is regime-dependent once costs are paid. Every number in this report is computed by tested code and reproduces from a committed snapshot via `make setup && make test && make run`; the source, tests and notebook accompany

this document.

A Software and reproducibility

The analysis is implemented in the `fxvol` Python package, organised into focused modules: `data` (snapshot load / live refresh), `preprocessing` (trading-day alignment), `indicators` (returns, rolling and EWMA volatility, SMA, RSI), `risk` (VaR/ES, drawdown, Sharpe/Sortino, Kupiec/Christoffersen), `models` (EWMA, GARCH, walk-forward forecasting, QLIKE, Diebold–Mariano), `diagnostics` (JB, ADF, KPSS, Ljung–Box, ARCH-LM), `backtest`, and `plots`. The notebook is a thin narrative layer that imports this package.

A `pytest` suite of 60 tests guards correctness: indicator golden values (e.g. Cutler RSI against a hand computation, the $\sqrt{252}$ annualization identity, the EWMA recursion), VaR/ES cross-checks (historical vs Normal on Gaussian data, Student- $t \rightarrow$ Normal as the degrees of freedom grow, $ES > VaR$), drawdown on a known path, Kupiec/Christoffersen calibration, a data-integrity check that no weekend row can enter the series, and a no-look-ahead assertion on the backtest. Continuous integration runs the linter, the test suite, and a headless execution of the notebook against the committed snapshot on every push. The full study reproduces offline with:

```
make setup && make test && make run
```

All inputs come from the committed Parquet snapshot; no network access is required to regenerate any number, table or figure in this report.

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